Arunabha Majumder

Mechanical | Mechatronics Engineer

Aalborg, Denmark | +45 71 62 24 00
somrkmv1997@gmail.com | 97arunabhasit027@gmail.com
Portfolio: arunabha22.github.io/arunabha-majumder | LinkedIn: arunabha-majumder-681264107



EDUCATION

PhD, Mechatronics and Robotics
Aalborg University | 2023 - present
Thesis expected March
Study abroad: NUS Singapore, 2025

MTech, Mechatronics
CSIR-CMERI, India | 2020 - 2022

BTech, Mechanical Engineering
SIT (VTU), India | 2015 - 2019

Senior Diploma in Fine Arts (Painting)
Bangiya Sangeet Parishad | 2008, Distinction

SKILLS

SolidWorks (CSWA) | 3D printing
Stiffness analysis in SolidWorks
Genetic algorithm optimisation
MATLAB/Simulink | LabVIEW | Arduino
Sensor integration | Test rigs
Qualisys motion capture and EMG
Hand sketching | Adobe Illustrator

AWARDS

Best Paper Award, IFToMM ISRM 2026
Best Major Project Award, 2019
TATA Technologies Innovation Grant

PUBLICATIONS

Hybrid actuated shoulder exoskeleton, ISRM 2026 (Springer)
Neural network-based control of a pneumatic muscle, IEEE CONECCT 2022

LANGUAGES

English (C1) | Bengali | Hindi

REFERENCES

Shaoping Bai, Professor
Aalborg University | sbh@mp.aau.dk

Lars Rosgaard Jensen, Associate Professor
Head of Section and Head of Labs, Aalborg University

Anderson de Souza Castelo Oliveira, Associate Professor
Materials and Production, Aalborg University

PROFILE

Mechanical and mechatronics engineer. I design, build and test robotic hardware and exoskeletons, from the first sketch to a tested prototype.

EXPERIENCE

Research Engineer, Shoulder Exoskeleton (PhD)
Aalborg University, Denmark | *2023 - Present*

- Designed a hybrid exoskeleton (spring + 6 Nm motor) that uses over 25% less energy. Best Paper Award.
- SolidWorks design and genetic algorithm optimisation; stiffness tested on a physical test setup.
- Integrated the motor, encoder and force sensors; built test rigs.
- Designed and modelled a novel 5-bar parallel robot with VSM; estimated its stiffness with less than 10% error and quantified human arm impedance.
- Collected EMG and motion data from 12 people; implemented an Arduino trigger to sync robot and Qualisys.
- Taught university courses and supervised Bachelor's students.

MTech Researcher, Pneumatic Muscles
CSIR-CMERI, India | *2020 - 2022*

- Fabricated and assembled pneumatic muscles and collected test data.
- Designed the outer PID loop of a neural network-based controller in MATLAB/Simulink and implemented it in LabVIEW to drive the muscle (0.3 - 0.78% error).

Bachelor's Project, Rehabilitation Exoskeleton
Siddaganga Institute of Technology, India | *2018 - 2019*

- Designed and built an upper-limb exoskeleton for a paralysed arm: CAD, machining, assembly and testing.